Path-Guided Reasoning (PGR) vs Flat Path Context

Query: "What connects Rational Grazing to Polyface Farm?" Ground-truth bridge: **Cows**

GRASP-RAG — Step-by-Step Chain (PGR)

Type: method **Degree:** 12
Rational Grazing is a systematic approach to managing pasture through controlled grazing periods that enhance productivity and pasture health. . .
[Source passage excerpt]

Keywords: livestock management, rotational grazing, animal health, pasture recovery
[Source passage: "Cows are moved across paddocks according to rational grazing schedules. . . "]

Type: livestock **Degree:** 8
Cows are central to regenerative farming systems. At Polyface Farm, Joel Salatin uses **Cows** as the primary animal in his rotational system. . .
[Source passage excerpt]

Keywords: rotational livestock, farm operations
[Source passage excerpt]

Type: organisation
Polyface Farm is Joel Salatin's regenerative farm in Virginia, known for multi-species rotational operations. . .

BER: HIT Answer names **Cows** at Step 3

PathRAG / LightRAG — Flat Retrieval

Rational Grazing is a grazing management method developed by André Voisin. It focuses on rest periods, optimal grazing time, and pasture productivity. . .

Polyface Farm, operated by Joel Salatin, is known for innovative sustainable farming. Salatin integrates multiple species in pasture-based systems. . .

Rational Grazing principles emphasise pasture recovery time, controlled stocking density, and mimicking natural herd migration patterns. . .

keywords: sustainable agriculture, grazing management, regenerative farming, land health

No step-by-step chain. No labeled bridge node. The LLM sees passages about each anchor — but no explicit path through the connecting entity.

✕ **BER: MISS** "...shared commitment to sustainable practices..." — **Cows never named**

Key difference: PGR places the bridge entity at a labelled step, making it structurally unavoidable. Flat retrieval gives the LLM correct passages but no path topology — the bridge node can be omitted without breaking the answer's surface fluency.